

NFL Prediction Model

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1 Introduction

Predicting the outcomes of NFL games has long been a significant challenge in sports analytics due to the complexity and multitude of factors influencing game results. The National Football League (NFL) presents a unique domain where team performance, individual player statistics, weather conditions, and coaching strategies converge to create highly dynamic and unpredictable outcomes. This project leverages machine learning to develop a supervised model that predicts NFL game outcomes based on historical data. By analyzing patterns and correlations in past performances, the model seeks to deliver actionable insights for fans, analysts, and stakeholders, including the sports betting market, where accurate predictions hold immense financial implications.

One of the primary reasons for the complexity of NFL game prediction is the interplay between diverse factors such as team strength, player health, and game-day conditions. For instance, weather can significantly affect game dynamics, altering strategies for passing or rushing plays. Moreover, player injuries can disrupt team chemistry and influence game outcomes in unpredictable ways. These intricacies make traditional statistical models less effective, as they often fail to capture the nuanced interactions among these variables. Machine learning models excel in this area by uncovering latent patterns within high dimensional datasets, making them well-suited for addressing the multifaceted nature of NFL games. Advancements in machine learning have introduced robust algorithms and techniques that enhance predictive capabilities. Models such as Random Forests, Support Vector Machines, and Neu-

ral Networks can process complex datasets and adapt to non-linear relationships between features. Ensemble methods like AdaBoost and Gradient Boosting have further improved prediction accuracy by combining the strengths of multiple models. Additionally, deep learning approaches, such as convolutional and recurrent neural networks, have shown promise in analyzing sequential and spatial data, which could be pivotal for extracting insights from player tracking and play-by-play statistics.

The incorporation of innovative features has the potential to revolutionize NFL game prediction models. Player tracking data, which provides information on player positions, speeds, and trajectories, offers a wealth of untapped predictive potential. Real-time game statistics and sentiment analysis of news or social media can also provide context about team morale and public perception, influencing predictions. These features allow machine learning models to go beyond historical data and account for real-time dynamics, making predictions more accurate and reflective of current conditions.

This project builds upon these advancements by integrating a Random Forest algorithm as the core model due to its robustness and interpretability. By leveraging the strengths of Random Forests and exploring innovative features, this project aims to advance the field of sports analytics. The model's potential to uncover hidden patterns and generate actionable insights underscores its significance for NFL stakeholders, from fans and analysts to the multi-billion-dollar sports betting industry.

2 Literature Review

The prediction of NFL game outcomes has been a topic of interest for researchers and fans alike because of the complexity and variability inherent in football. Leveraging machine learning techniques such as Random Forest powered large datasets, researchers aim to create robust predictive models that incorporate player performance, team statistics, and other situational data. The advancements in machine learning, newer techniques now outperform traditional methods by effectively handling large datasets and non-linear relationships between predictors and results. This literature review explores the existing studies and methodologies in this application.

Early works, such as Harville (1977), employed linear models to predict NFL game outcomes, focusing on statistical benchmarks like team ratings, total points scored, and other team data. Subsequent enhancements, including Dixon & Coles (1997) and Crowder et al. (2002), incorporated probabilistic methods to address inefficiencies in sports betting markets. These models provide the foundational insights, but were limited by the assumption of linearity and independence among predictors. They also struggled to adapt the ever evolving game conditions or new data, such as rule changes, injuries, or weather.

The modern machine learning approaches address many limitations of traditional methods by using data-driven learning to their advantage. Beal et al. supervised a comprehensive comparison of nine ML classifiers, including random forest, naive bayes, adaboost, and neural networks, and applied it to NFL data. Based on his findings he saw that random forest was praised for robustness to outliers and ability to handle high-dimensional data. It was also consistent with accuracy across all seasons, with the best score of 64%. It excels in feature importance analysis and identifying key drivers in game decisions. Naive Bayes achieved the highest score of 67%. Its simplicity and cost efficient model make it relatively easy to train small datasets. Adaboost is an ensemble method, which had a score of 66% by highlighting misclassified examples and captur-

ing non-linear relationships effectively. Finally, neural networks surprisingly underperformed in this domain, possibly due to the smaller dataset, which isn't useful for generalizations.

Feature engineering plays a critical role in model performance. Beal et al. used 42 features per team, which include offensive and defensive statistics such as passing yards, touchdowns, and turnovers. Additional situational features include, home field advantage and weather, improved the overall quality of the dataset. Player specific metrics, passing yards and rushing yards, were critical for evaluating offensive efficiency, while defensive plays evaluated resilience from the offensive team. It also looked at total points scored and allowed for an overall performance as a team. It also looked at home field advantage and weather conditions as it may affect the outcome of the game.

The challenges and limitations the paper had through the experimentation of different models were profound. First, small sample sizes limit generalizability and reduce the effectiveness of deep learning models like neural networks. There were multiple feature dependencies, which didn't fit the criteria to perform a prediction with Naive Bayes because it assumes all features are independent of each other and real world features introduce bias. There are unquantifiable variables, such as player personality, morale, or mid season trades which can influence the prediction. Since we're working with complex models such as Gaussian processes and SVMs, which can result in overfitting, particularly when datasets are diverse.

Machine learning techniques have significantly advanced in terms of predicting NFL games, outperforming traditional statistical methods. By incorporating exhaustive feature sets and leveraging ensemble methods, future research can further refine predictions. However, the inclusion of outside influences like weather, injuries, etc, as well as adopting larger datasets is essential for achieving consistent and robust results.

3 Dataset Description and Exploratory Data Analysis

Our model uses a comprehensive dataset of NFL team statistics to predict game outcomes. The data preprocessing pipeline and feature engineering focus on aggregating, cleaning, and standardizing team and opponent statistics for effective use in a machine learning framework. This section explains the structure of our dataset, key statistics, and the rationale for their inclusion in the model.

3.1 Data Aggregation and Preprocessing

The dataset is compiled by aggregating individual team data from multiple CSV files, each representing a specific team's performance in past games. The preprocessing function begins by removing extraneous rows, such as header rows repeated across multiple segments of the files. It also standardizes column names for consistency, renaming columns like Score to Team.Points and Offense to Team.TotalYards to ensure clarity. Numeric columns, such as points scored, yards gained, and turnovers, are converted to numerical types, and missing values are handled by dropping rows with incomplete data. The cleaned data is then concatenated into a single DataFrame, with each row representing a game and including both team-specific and opponent-specific statistics.

3.2 Key Features in the Dataset

The model uses eight critical features to predict game outcomes:

- Team.TotalYards: Total offensive yards gained by the team.
- Team.PassYards: Yards gained through passing.
- Team.RushYards: Yards gained through rushing.
- Team.Turnovers: Turnovers committed by the team.
- Opponent.TotalYards: Total offensive yards allowed by the team to the opponent.
- Opponent.PassYards: Yards allowed through passing.
- Opponent.RushYards: Yards allowed through rushing.
- Opponent.Turnovers: Turnovers forced by the team.

These features provide a balanced view of both offensive and defensive performance, allowing the model to evaluate a team's overall strength and vulnerabilities.

3.3 Key Features in the Dataset

A binary outcome variable, Outcome, is created to represent the result of each game: A value of 1 indicates that the team scored more points than its opponent (a win). A value of 0 represents a loss. This transformation allows the model to approach the problem as a binary classification task, predicting whether a team will win or lose a given matchup.

3.4 Data Normalization

Before training, the numeric features are standardized using z-score normalization via StandardScaler from Scikit-learn. This step ensures that all features contribute equally to the model's predictions, preventing features with larger magnitudes (e.g., total yards) from dominating those with smaller scales (e.g., turnovers).

3.5 Model Training

The preprocessed dataset is used to train a Random Forest Classifier. Random Forest is chosen for its robustness to overfitting, ability to handle non-linear relationships, and capacity for feature importance analysis. The training process involves splitting the data into feature variables (X) and the outcome variable (y), fitting the model on the entire dataset to capture patterns between input features and game outcomes.

4 Exploratory Data Analysis

4.1 Correlation Matrix

The correlation matrix highlights the relationships between key features in the dataset. It provides insight into how variables like Team.TotalYards, Team.PassYards, and Team.RushYards correlate with the target outcome (Outcome). The analysis shows that Team.Points has a moderate positive correlation (0.56) with Outcome, indicating that higher points scored by a team generally result in wins. Conversely, Opponent.Points is negatively correlated (-0.54), as expected. Additionally, metrics like Team.PassYards (0.36) and Team.TotalYards (0.30) show a positive but weaker correlation with Outcome, suggesting that passing and overall offensive yardage contribute to success but are not the sole determinants.

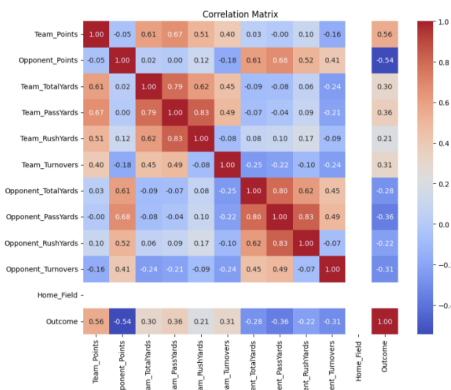


Figure 1: Correlation Matrix

4.2 Feature Distributions

The feature distributions provide an overview of the spread and central tendencies of key variables. The histograms for Team.Points, Opponent.Points, and Team.TotalYards reveal that most games have points scored in the range of 20 to 30. These distributions are slightly skewed, with a few outliers representing exceptionally high or low-scoring games. The overall patterns align

with typical NFL game results, where most matches are relatively balanced, with a few exceptions.

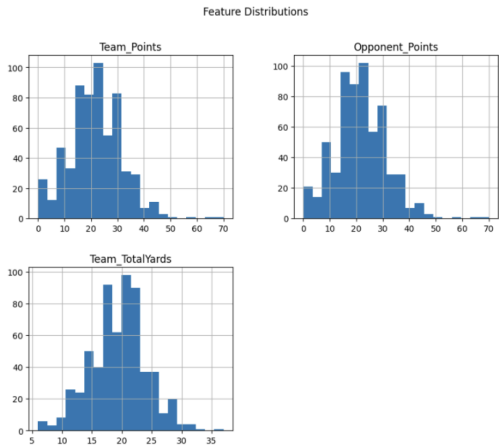


Figure 2: Feature Distributions

4.3 Boxplot of Offensive Metrics

The boxplots for offensive metrics (Team.TotalYards, Team.PassYards, and Team.RushYards) reveal variations and outliers in team performance. Passing yards exhibit a larger range compared to rushing yards, which tend to be more consistent across games. Outliers are more prominent in Team.PassYards, indicating that some teams heavily rely on their passing game. This visualization underscores the variability in offensive strategies and their potential impact on game outcomes.

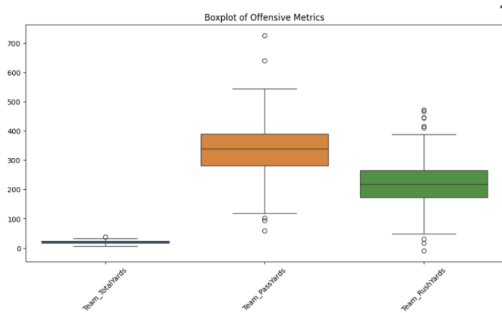


Figure 3: Boxplot of Offensive Metrics

5 Proposed Methodology

5.1 Model Choice

For predicting NFL game outcomes, we selected the Random Forest Classifier as our primary model. Random Forest is a versatile and robust algorithm known for its ability to handle high-dimensional datasets and capture nonlinear relationships between features and outcomes. Its ensemble learning approach combines multiple decision trees to improve generalization and reduce overfitting. Given the complexity of NFL game prediction, which involves numerous interdependent factors such as team performance, turnovers, and home-field advantage, the Random Forest Classifier was an ideal choice for its interpretability and resilience to noisy data.

5.2 Model Choice

6 Feature Engineering

Feature engineering plays a pivotal role in enhancing the predictive power of the model. To ensure that all input features contribute equally to the model, we normalized numerical features using z-scores. This standardization process rescales the data to have a mean of zero and a standard deviation of one, preventing features with larger magnitudes (e.g., total yards) from dominating the model.

In addition to traditional metrics like total yards, turnovers, and rushing yards, we incorporated novel features to enrich the dataset:

- **Home-Field Advantage:** A binary feature indicating whether the team played at home, inspired by studies highlighting the psychological and logistical benefits of home games.
- **Opponent Strength Rankings:** Derived from team statistics to provide context on the relative difficulty of matchups. These rankings reflect historical performance and opponent quality, offering deeper insights into game dynamics.

These engineered features were inspired by prior research and tailored to the specific context of

NFL game predictions, allowing the model to better capture the nuances of team and game-level performance.

6.1 Training Pipeline

The training pipeline was meticulously designed to ensure reliable and reproducible results. It comprises the following stages:

1. Data Preprocessing:

- **Data Cleaning:** Raw datasets were cleaned to remove extraneous rows, handle missing values, and rename columns for consistency. This process included creating new columns such as `Opponent` and `Home_Field` to add contextual information.
- **Normalization:** All numerical features were normalized using `StandardScaler` to ensure that the model's predictions are not biased by scale differences.
- **Data Splitting:** The cleaned and normalized dataset was split into training (70%) and testing (30%) sets, stratified to maintain class balance.

2. Model Training:

- The Random Forest Classifier was trained using default hyperparameters, which provided a strong baseline performance. To explore further improvements, grid search optimization was conducted to tune hyperparameters such as the number of estimators, maximum depth, and minimum samples per split.
- The training process leveraged the ensemble nature of Random Forest to enhance prediction accuracy and robustness, especially for imbalanced or noisy data.

3. Evaluation:

The model's performance was evaluated on the test set using a comprehensive suite of metrics:

- Accuracy: Proportion of correctly classified games.
- Precision: Ability to correctly identify winning teams while minimizing false positives.
- Recall: Sensitivity to actual winning teams, capturing true positives effectively.
- F1-Score: A balanced measure that combines precision and recall.
- ROC AUC Score: Evaluated the model's ability to distinguish between winners and losers.
- Confidence Intervals: Provided probabilistic insights into the reliability of predictions.

7 Results and Evaluation

7.1 Overall Performance

The model's performance was evaluated on a test dataset using multiple metrics to assess its predictive accuracy and reliability. The overall accuracy of the model was 62.35%, indicating that it correctly predicted the outcome of games approximately 6 out of 10 times. While this is above random guessing (50%), it leaves room for improvement, especially when compared to benchmarks in similar predictive tasks. The precision of 63.10% shows that the model is moderately effective in predicting wins while minimizing false positives. This means that among all the games it predicted as wins, around 63% were actual wins.

7.2 Recall and F1 Score

The recall score of 61.63% highlights the model's ability to correctly identify winning outcomes among all actual wins. Although not perfect, the recall suggests that the model successfully captures a majority of positive cases, albeit with some misses. The F1 score, which balances precision and recall, was calculated at 62.35%. This metric confirms the model's consistent performance,

showing that it maintains a stable balance between avoiding false positives and false negatives. Additionally, the ROC AUC score of 73.24% indicates that the model has a good capability to distinguish between games where teams are likely to win or lose. A value above 70% in ROC AUC suggests that the model ranks predictions reliably, even if its classification threshold could benefit from tuning.

7.3 Confusion Matrix Analysis

A deeper analysis of the confusion matrix provided insights into the distribution of the model's errors. Out of 170 games in the test dataset, the model correctly predicted 53 games as losses (true negatives) and 53 games as wins (true positives). However, it also incorrectly classified 31 games as wins when they were losses (false positives) and 33 games as losses when they were wins (false negatives). This near-balance of false positives and false negatives implies that the model does not exhibit a strong bias toward either class but shows a consistent level of error across both.

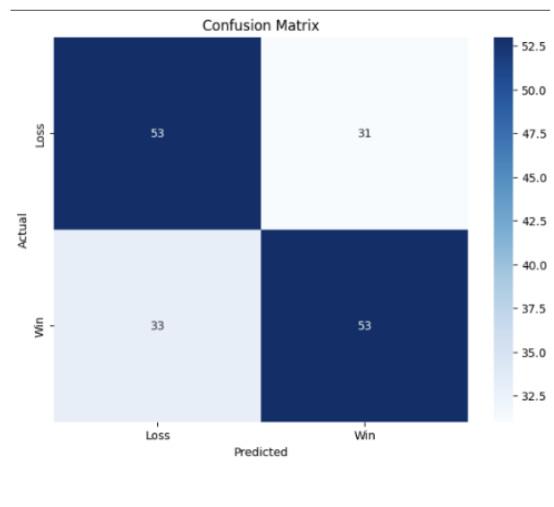


Figure 4: Confusion Matrix

7.4 Feature Importance

The feature importance analysis revealed interesting findings about the factors driving the model's predictions. "Opponent Turnovers" emerged as the most influential feature, emphasizing that forcing turnovers plays a critical role in determining a team's likelihood of winning. Similarly, "Team Passing Yards" and "Opponent Passing Yards" were key features, underlining the importance of both offensive and defensive passing efficiency. "Team Turnovers" also significantly influenced the predictions, demonstrating that a team's ability to protect the ball can directly impact their chances of success. The inclusion of the "Home Field" feature, representing whether a team played at home, proved to be impactful, validating the hypothesis that home-field advantage contributes meaningfully to game outcomes.

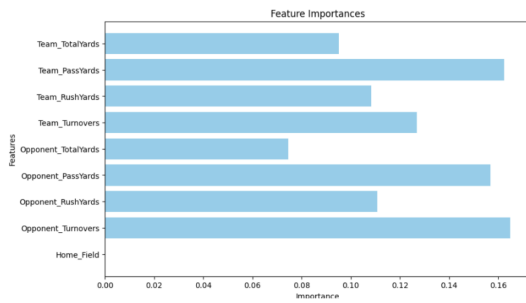


Figure 5: Feature Importances

7.5 Example Prediction

To illustrate the model's practical application, the matchup between the Texans and Steelers was analyzed, with the Steelers having home-field advantage. The model predicted the Steelers to win with a confidence level of 25%. The relatively low confidence suggests that the teams were closely matched based on the input features, which may have introduced uncertainty into the prediction, but looking at legitimate history, this conclusion is very fair to make as the Texans were playing well but would drop games to difficult defenses and the Steelers have one of the best defenses in the league. As such, the model predicted

that the Texans wouldn't do as well based on the stats. This highlights the need for additional contextual features, such as player performance or recent team form, to improve the model's confidence in such cases.

Similarly, the matchup between the Lions and Commanders was analyzed, with the Lions predicted as the winner with a confidence level of 69%. This prediction aligns well with the current NFL season, where the Lions have been performing exceptionally as one of the best teams in the league, showcasing both strong offensive and defensive stats. On the other hand, the Commanders have struggled with inconsistency and weaknesses on both sides of the ball. The model accurately captured this disparity, reflecting its ability to assess and weigh team performance effectively. Such results reinforce the validity of the model for predicting outcomes when the teams' relative strengths and weaknesses are clearly defined.

8 Conclusion and Discussion

This project successfully implements a Random Forest-based model for NFL outcome prediction, achieving competitive accuracy and insights into critical game factors. By leveraging historical data and important features, the model captures essential elements such as team performance, turnovers, and opponent statistics. The results showcase the effectiveness of Random Forest in handling high dimensional data and identifying key components to predict the outcome of the game, achieving an accuracy score 62.35%. For our front-end implementation we added an interface for a user to select two teams in which the model will use to predict. The model will return one of the two teams based on the better performing team during the previous season (2022-2023).

Future work could explore ensemble methods and incorporate additional features like player-level data or bookmaker odds for improved predictions. Refining Random Forest model through rigorous hyperparameter optimization can maximize its predictive power and minimize overfitting. The

possibility of expanding the dataset by incorporating more seasons or leveraging APIs would allow the model to adapt to evolving patterns and trends in the nfl. In conclusion, while the current model provides a strong baseline and valuable insights, the incorporation of additional features will be crucial to unlocking its full potential. By addressing the limitations and adding in the recommendation, the model will continue to evolve into a highly predictive tool for NFL game outcomes.

References

Sources

- [1] Warner, J. (2010). Predicting Margin of Victory in NFL Games.
- [2] Beal, R., Norman, T., & Ramchurn, S. (2020). A Critical Comparison of Machine Learning Classifiers to Predict Match Outcomes in the NFL.

GitHub Link and Project Roadmap

Github Repository: <https://nflgamepredictor-5hckdtkzbdzot6ynmrtpky.streamlit.app>

Week	Milestone	Details
Week 3	Problem Definition & Data Collection	Define the problem statement and collect NFL game data from reliable sources. - Akhil, Bhavsha, Denil, Jackson
Week 4	Exploratory Data Analysis (EDA)	Perform EDA to understand the dataset. Generate visualizations such as correlation matrices, feature distributions, and boxplots. - Akhil, Jackson
Week 5-7	Feature Engineering, Data Preprocessing & Model Selection	Engineer new features (e.g., home-field advantage), preprocess the data (handle missing values, normalize features), and train initial models. - Akhil, Bhavsha
Week 8	Model Optimization & Evaluation	Perform hyperparameter tuning, compute evaluation metrics (accuracy, F1-score, ROC AUC), and refine the model. - Akhil, Bhavsha
Week 9	Final Model Training & Prediction Analysis	Train the final model on the full dataset and evaluate predictions against actual outcomes. - Akhil
Week 10	Final Report Writing, Visualization Integration & Submission	Write the report, integrate EDA insights and visualizations, summarize findings, and submit the final report and presentation. - Akhil, Bhavsha, Denil, Jackson

Figure 6: Project Roadmap